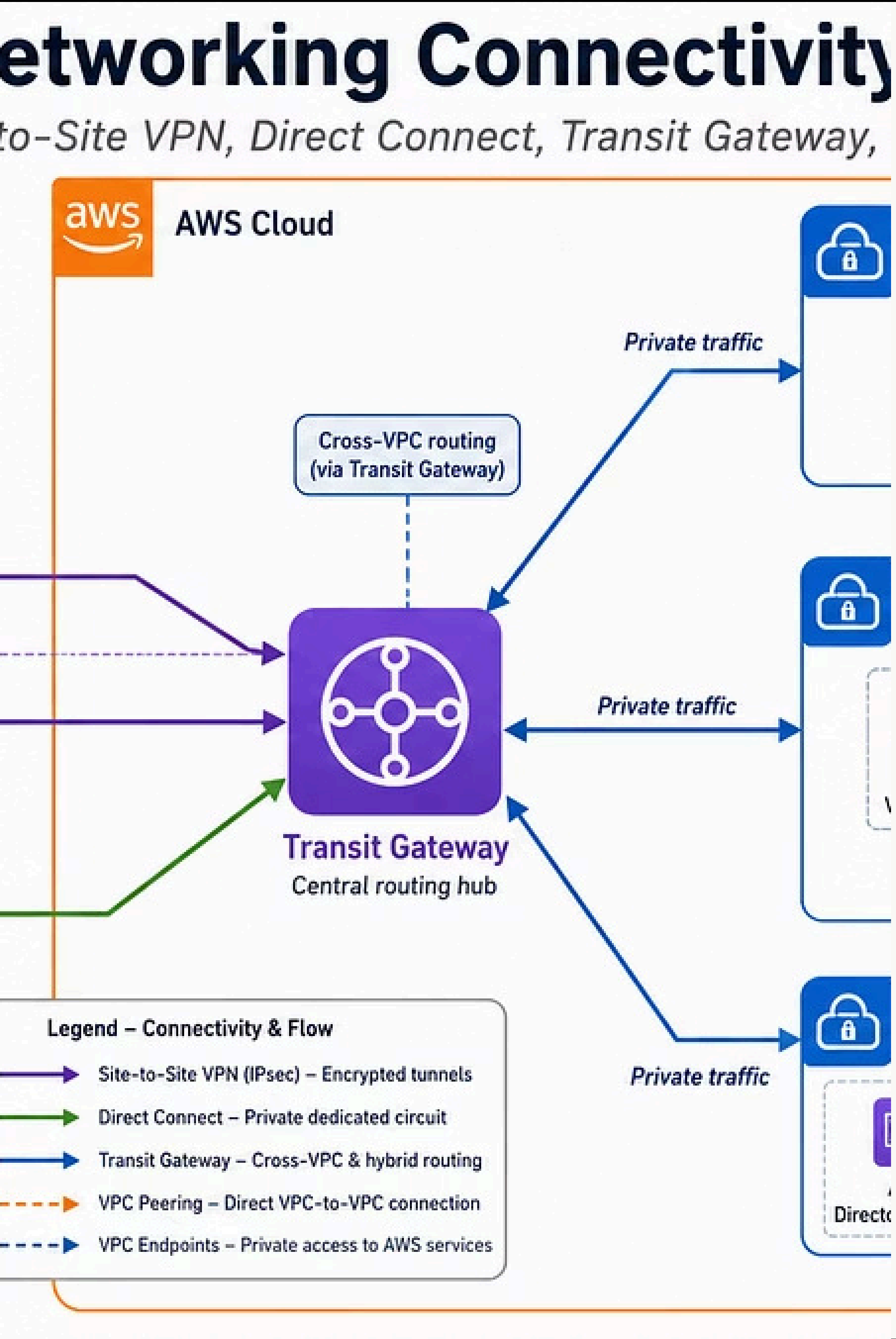


BASESTACK ACADEMY · AWS CLOUD ACCELERATOR · COHORT 1 · BEGINNER-FRIENDLY · SAA-C03 ALIGNED

VPC Peering & Connectivity

WEEK 3 · DAY 5

NETWORKING & VPC

Today we cover all five AWS connectivity tools — VPC Peering, Site-to-Site VPN, Direct Connect, Transit Gateway, and VPC Endpoints — when to use each one, and how to answer the exam question: **"Which connectivity option is MOST appropriate for this scenario?"**

Overview — Why VPCs Need to Connect

Real-world AWS deployments rarely live in a single VPC. Most organizations run separate VPCs for production, development, staging, and different business units — for security isolation and cost control. But those VPCs eventually need to communicate with each other, reach on-premises offices, or access AWS services privately. AWS gives you **five tools** to do this. Choosing the wrong one is one of the most expensive cloud architecture mistakes — both in cost and in exam marks.

Why This Matters for SAA-C03

Domain 1 and Domain 3 both test connectivity. **"How do you connect VPC A to VPC B?"** and **"How do you connect your data center to AWS?"** appear in multiple questions every exam attempt.

Nigerian banks like Zenith, GTBank, and UBA — separating core banking from analytics from internal tools — all face this exact problem. Knowing when to use Transit Gateway vs VPC Peering vs Direct Connect vs VPN is the mark of a mid-level to senior cloud engineer.

🧠 Quick Mental Model — Road Analogy

→ **VPC Peering**

Private road directly between two specific offices — no public traffic, but only connects those two buildings.

→ **Site-to-Site VPN**

Armored van on a public highway — encrypted tunnel, but still traveling on shared infrastructure with variable latency.

→ **Direct Connect**

Dedicated private motorway with no public traffic — consistent, fast, but requires physical construction to build.

→ **Transit Gateway**

Central motorway interchange — many roads, one hub. Every VPC connects once, and the hub handles all routing.

→ **VPC Endpoints**

Internal office lift to AWS — never exits the building. Private access to AWS services without touching the internet.

VPC Peering — Private, Direct & Non-Transitive

VPC Peering creates a direct, private network link between two VPCs using AWS's private backbone — never the internet. Traffic between peered VPCs behaves as if the instances are on the same private network. It works across AWS accounts and across regions, but comes with one critical limitation you must never forget on the exam.

How It Works

A peering connection (`pcx-xxxxx`) is created between two VPCs. EC2 in VPC A (10.0.1.5) can communicate directly with RDS in VPC B (172.16.1.8) over the AWS private backbone. No internet exposure. No NAT Gateway needed. No encryption needed — the traffic is already private.

📋 Route Tables — BOTH VPCs

Peering alone does **NOT** route traffic. You must add a route in VPC A's route table pointing to VPC B's CIDR, *and* a route in VPC B's route table pointing to VPC A's CIDR. Both must be done. One-way routes = one-way traffic only.

🛡️ Security Group Referencing

In the same region, a Security Group in VPC A can reference a Security Group in VPC B as a source — more secure than using CIDR blocks because it follows the resource, not the IP.

🚫 Key Rules

No CIDR Overlap

VPC A (10.0.0.0/16) and VPC B (10.0.0.0/24) **cannot peer** — overlapping ranges. Plan your IP addressing before creating VPCs.

⚠️ Non-Transitive (Critical!)

If A↔️B and B↔️C, then **A CANNOT reach C through B.** Traffic does NOT transit through a peered VPC. For 3 VPCs: need A-B + B-C + A-C (3 connections). For 10 VPCs: 45 connections → use Transit Gateway instead.

🌏 Cross-Account & Cross-Region

Peering works across AWS accounts and across regions. Inter-region peering has higher data transfer costs than intra-region peering.

Connecting On-Premises to AWS — VPN vs Direct Connect

Both tools connect your physical office or data center to AWS — but they differ dramatically in cost, performance, setup time, and use case. Knowing which to recommend is a core SAA-C03 skill and a real-world architecture decision that affects both budget and reliability.

🔒 AWS Site-to-Site VPN

Encrypted tunnel over the public internet

- **How:** IPsec tunnel between your on-premises router (Customer Gateway) and a Virtual Private Gateway (VGW) in your VPC
- **Encryption:** YES — IPsec encrypted by default. Secure out of the box.
- **Speed:** Up to 1.25 Gbps per VPN connection. Latency varies because it uses the public internet.
- **Redundancy:** AWS creates 2 tunnels automatically. Both must be monitored — no auto-failover without configuration.
- **Setup time:** Minutes. No physical provisioning needed.
- **Cost:** ~$0.05/hr per connection + data transfer charges. Affordable for low-to-medium traffic volumes.
- **Best for:** Branch offices, dev environments, backup for Direct Connect, organizations not ready for a DX commitment.

⚡ AWS Direct Connect

Dedicated private fiber — never touches the internet

- **How:** Physical dedicated fiber from your data center to an AWS Direct Connect location. 100% private path.
- **Encryption:** NOT encrypted by default. Add a VPN over DX for encryption + private path combined.
- **Speed:** 1 Gbps, 10 Gbps, or 100 Gbps dedicated. Consistent, predictable low latency — no internet variability.
- **Redundancy:** Provision two DX connections in different DX locations. Use VPN as automatic backup if DX fails.
- **Setup time:** Weeks to months. Requires physical provisioning through an AWS Direct Connect partner.
- **Cost:** Port hour fee (~$216/month for 1Gbps) + significantly reduced data transfer rates. Saves money at high volume.
- **Best for:** High-bandwidth, low-latency, compliance-sensitive workloads with predictable traffic patterns.

💡 **Pro Tip — Best of Both Worlds:** For CBN-regulated environments requiring both a private path AND encryption, combine Direct Connect (private, low-latency) with a Site-to-Site VPN layered on top (IPsec encryption). This satisfies both performance and security compliance requirements simultaneously.

Transit Gateway & VPC Endpoints — Scale & Private Access

🌐 Transit Gateway — Hub-and-Spoke for Many VPCs

When you have many VPCs that need to communicate, VPC Peering creates an unmanageable mesh. Transit Gateway is a regional hub — each VPC and on-premises network connects **once** to the TGW, and TGW handles all routing between attachments.

- **Attachments:** VPCs, Site-to-Site VPNs, Direct Connect Gateways, and other TGWs (for inter-region connectivity)
- **Route Table Isolation:** Separate TGW route tables enforce isolation — Dev VPC cannot reach Prod even through TGW
- **vs Peering:** 10 VPCs with peering = 45 connections. With TGW = 10 attachments. Scales cleanly.
- **Cost:** \$0.05/hr per attachment + \$0.02/GB data processed. Regional resource — must set up per region.

🔌 VPC Endpoints — Private Access to AWS Services

VPC Endpoints let your VPC resources access AWS services without traversing the internet — no NAT Gateway, no internet gateway, no public IP required. Two types exist:

Gateway Endpoints — FREE

For **S3 and DynamoDB only.** Adds a route table entry pointing S3/DynamoDB traffic to the endpoint over the AWS private backbone. Cost: **\$0/hr. \$0/GB.** There is literally no reason not to use this for S3 and DynamoDB in private subnets.

Interface Endpoints — Paid (PrivateLink)

For **all other AWS services** — SSM, SQS, SNS, ECR, Secrets Manager, and 100+ more. Creates an Elastic Network Interface (ENI) with a private IP in your subnet. Cost: ~\$0.01/hr per AZ + ~\$0.01/GB processed. Powered by **AWS PrivateLink**, which also allows SaaS vendors to expose services privately to your VPC.

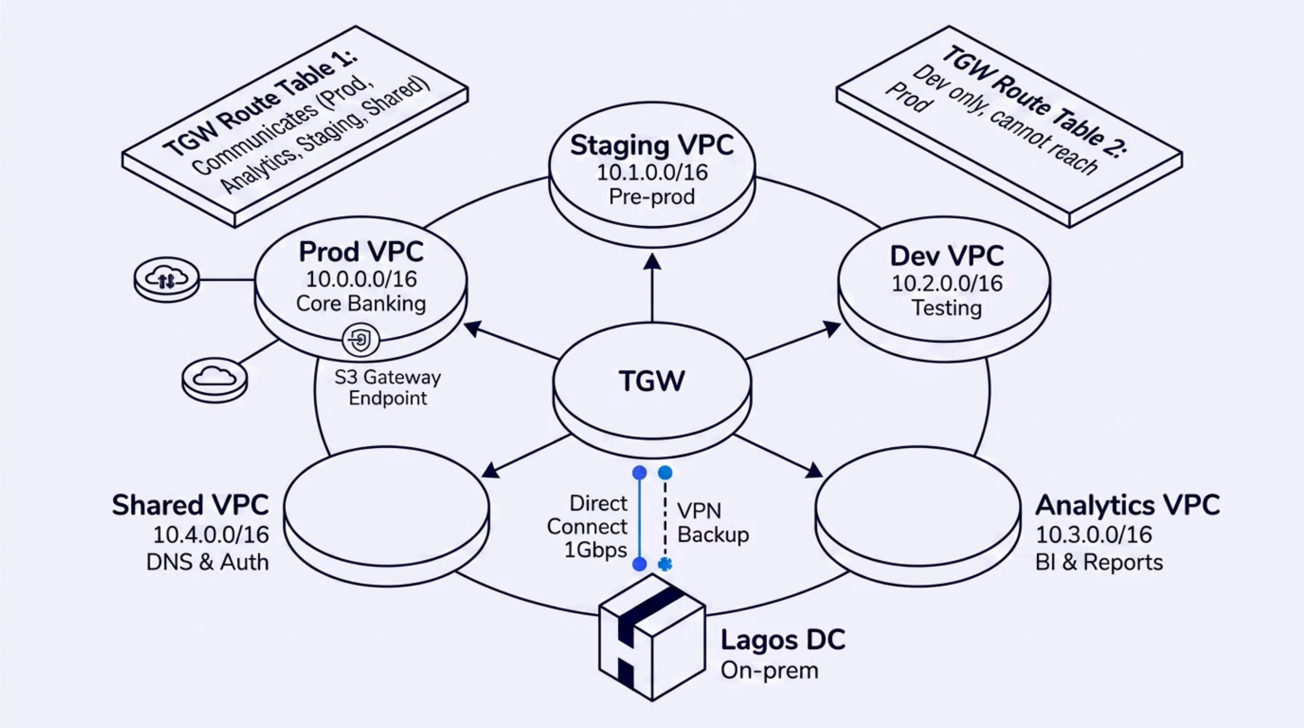
Decision Framework — Which Connectivity Tool Do You Choose?

Match the scenario keyword to the right tool. This framework answers most SAA-C03 connectivity questions. The exam will describe a business requirement — your job is to map the keywords to the correct service. Use this table as your mental lookup during the exam.

Scenario / Keywords	Best Tool	Why	Avoid
Connect 2–3 VPCs, same or different accounts, simple topology	**VPC Peering**	Simple, cheap, private, low overhead. Works across accounts and regions.	TGW — overkill for 2–3 VPCs
Connect 10+ VPCs, need traffic isolation between VPCs, hub-and-spoke	**Transit Gateway**	Hub-and-spoke scales cleanly. Route table isolation enforces security boundaries. 10 attachments vs 45 peering connections.	Peering — mesh becomes unmanageable
On-premises office → AWS, low volume, quick setup needed	**Site-to-Site VPN**	Fast to provision (minutes), IPsec encrypted by default, affordable. No physical provisioning required.	DX — weeks lead time, higher cost
On-premises data center → AWS, high bandwidth, low latency, high volume	**Direct Connect**	Dedicated, consistent, cheap per-GB at scale. No internet variability. Predictable latency for compliance workloads.	VPN — 1.25 Gbps limit, variable latency
Need encryption AND dedicated private connection (CBN compliance)	**Direct Connect + VPN**	DX gives private path. VPN layered on top adds IPsec encryption. Best of both worlds for regulated environments.	DX alone — not encrypted by default
Private subnet EC2 needs to reach S3 or DynamoDB without internet	**S3/DynamoDB Gateway Endpoint**	FREE. Adds route table entry. Traffic stays on AWS backbone. No NAT Gateway needed. Zero cost, zero reason not to.	NAT Gateway — costs money, exposes to internet
Private subnet EC2 needs to reach other AWS services (SSM, SQS, etc.)	**Interface Endpoint (PrivateLink)**	Creates ENI with private IP in your subnet. Paid (~$0.01/hr per AZ). Enables private access to 100+ AWS services.	NAT Gateway — unnecessary cost and internet exposure

Architecture — Enterprise Multi-VPC Connectivity

A complete enterprise connectivity model using **Transit Gateway** as the central hub, **Direct Connect** for the data center, and **VPC Endpoints** for private AWS service access. This is the reference architecture for a regulated financial institution operating across multiple AWS accounts and on-premises locations.



TGW Route Table Isolation

Prod + Analytics + Staging + Shared VPCs are in one TGW route table — they can all communicate freely. Dev VPC is in a **separate TGW route table** — it can reach Shared Services (DNS/Auth) but **cannot reach Prod** even through the Transit Gateway. This enforces security boundaries at the network level, not just with Security Groups.

Key Design Decisions

- **Lagos Data Center:** Direct Connect 1Gbps for high-volume, low-latency traffic to Analytics VPC. VPN configured as automatic failover.
- **S3 Gateway Endpoint in Prod VPC:** Core banking EC2s in private subnets write audit logs to S3 via AWS backbone — zero internet exposure, zero NAT cost.
- **Shared Services VPC:** Centralized DNS and authentication accessible to all VPCs — reduces duplication and simplifies management.
- **Scalability:** Adding a new VPC requires only one TGW attachment — no re-architecture of existing connections.

Real-World Scenario — Zenith Bank-Style Cloud Expansion

A tier-1 Nigerian bank operates three AWS accounts — Production (core banking), Analytics (BI), and Development — plus two physical data centers in Lagos and Abuja. The cloud architect must design a connectivity architecture satisfying **CBN security requirements** for isolation, encryption, and auditability.

Transit Gateway Connects All AWS Accounts

3 AWS VPCs (Prod, Analytics, Dev) attach to TGW. Separate TGW route tables: **prod-rt** (Prod + Analytics + Shared Services) and **dev-rt** (Dev + Shared Services only). Dev **cannot** reach Prod even through TGW. Avoids 3 peering connections today and is ready for 10 more VPCs as product lines expand.

✅ **Result:** All VPCs reach Shared Services. Prod↔️Analytics for real-time reporting. Dev is isolated. Scales to future VPCs with no re-architecture.

Direct Connect — Lagos Data Center

The Lagos DC sends **500GB+ daily** to Analytics VPC for reconciliation. Treasury systems need <20ms latency for real-time position updates — internet VPN latency of 80–200ms is unacceptable for trading systems.

✅ **1Gbps DX provisioned** via an AWS partner in Lagos. Data transfer cost reduced 70%. Treasury latency: 15–25ms consistent. Added Site-to-Site VPN as automatic failover — if DX fails, traffic routes through VPN (higher latency but maintained connectivity).

Site-to-Site VPN — Abuja Data Center

Abuja handles disaster recovery and regulatory archiving. Data volume: 10–50GB/day. No strict latency requirements. DX to Abuja would cost ~$216/month port fee plus 2–4 week provisioning — overkill for this use case.

✅ **VPN setup in under 1 hour.** Monthly cost: ~$36 (vs $216+ for DX). Two tunnels configured and both monitored. CBN-compliant encrypted connectivity. Abuja VPN also serves as backup path if Lagos DX fails.

S3 Gateway Endpoints — Core Banking Audit Logs

Core banking EC2s run in private subnets with **NO NAT Gateway route** (CBN requires zero internet path from core banking systems). These servers must write encrypted audit logs to S3. Without an endpoint — no path to S3 exists.

✅ **S3 Gateway Endpoints added** to Prod VPC private subnet route tables. Traffic routes to S3 via AWS backbone — never the internet. No NAT Gateway needed. No internet route created. Cost: $0. CBN isolation requirement fully maintained.

Exam Traps + SAA-C03 Practice Question

⚠️ Top 5 Exam Traps

1 Peering is NON-TRANSITIVE

A↔️B↔️C does **NOT** mean A can reach C. Each path requires its own connection. For many VPCs → Transit Gateway. This is the most-tested VPC Peering concept on SAA-C03.

2 Route Tables — BOTH VPCs

Peering alone routes nothing. Update VPC A's route table *and* VPC B's route table. One-way routes = one-way traffic only. The exam will describe a one-way connectivity issue — the answer is always "add the missing route."

3 Direct Connect is NOT Encrypted

DX = private, not encrypted. Add VPN over DX for encryption + private path combined. CBN and similar regulations often require both — this combination is a common exam scenario.

4 VPN Has TWO Tunnels — Monitor Both

AWS provisions 2 tunnels automatically. If one fails with no failover configuration → outage. Both must be active and monitored. The exam may describe a partial VPN failure scenario.

5 Gateway Endpoints are FREE

For S3 and DynamoDB, Gateway Endpoints cost nothing and remove the need for a NAT Gateway. Always use them in private subnets. If the exam mentions private subnet → S3 access, Gateway Endpoint is almost always the answer.

📝 Practice Question — Select TWO

A Lagos fintech has five VPCs and a Lagos data center. **ALL VPCs must reach a Shared Services VPC.** The data center sends **300GB/day** to AWS. The Development VPC must **NOT** reach Production. Which configurations meet ALL requirements?

A — Full-mesh VPC Peering + Direct Connect

× Wrong. Full-mesh = 10 connections. Cannot enforce Dev↔️Prod isolation with peering alone.

B — Transit Gateway + Direct Connect ✓

✅ **Correct.** TGW handles all VPCs with isolation via separate route tables. DX suits high-volume (300GB/day) with consistent latency.

C — Transit Gateway + Site-to-Site VPN

× Partially correct. TGW is right. But VPN at 300GB/day becomes expensive and slow vs DX.

D — VPC Peering to Shared Services + Direct Connect

× Wrong. 5 peering connections don't scale. Cannot enforce Dev↔️Prod isolation cleanly.

Answer: only B fully satisfies all requirements.

Key Takeaways & Resources — Week 3, Day 5

✅ What You Learned Today

VPC Peering

Private, direct, **non-transitive** — update BOTH route tables. No CIDR overlap allowed. Works across accounts and regions. For 2–3 VPCs only.

Transit Gateway

Hub-and-spoke architecture. Route table isolation enforces security boundaries. Scales cleanly — 10 attachments vs 45 peering connections for 10 VPCs.

Site-to-Site VPN

Encrypted IPsec over the internet. VGW + CGW + 2 tunnels (monitor both). Fast setup, affordable. Best for low-to-medium volume on-premises connectivity.

Direct Connect

Dedicated private fiber — NOT encrypted by default. 1/10/100 Gbps. Consistent low latency. DX + VPN = private path + encryption (best of both for compliance).

VPC Endpoints

Gateway Endpoints: **FREE** for S3 + DynamoDB. No internet, no NAT needed. Interface Endpoints: paid, all other services, powered by PrivateLink. Client VPN = individual laptops; Site-to-Site VPN = office-to-AWS.

> 🎯 **Decision Shortcut:** Few VPCs → Peering · Many VPCs → TGW · On-prem low volume → VPN · On-prem high volume → DX · Private subnet → S3 → Gateway Endpoint

📚 Key Resources

01

VPC Peering User Guide

docs.aws.amazon.com/vpc/latest/peering/

02

Transit Gateway User Guide

docs.aws.amazon.com/vpc/latest/tgw/

03

AWS Direct Connect Features

aws.amazon.com/directconnect/features/

04

VPC Endpoints & PrivateLink

docs.aws.amazon.com/vpc/latest/privatelink/

05

SAA-C03 Exam Guide (Domain 3)

aws.amazon.com/certification/certified-solutions-architect-associate/

📋 Before Next Class

1. Console lab: create 2 VPCs, peer them, update BOTH route tables, test EC2-to-EC2 connectivity
2. Create an S3 Gateway Endpoint — confirm private subnet can reach S3 without a NAT Gateway
3. Sketch the Zenith Bank scenario on paper — label which tool connects what to what
4. Recall the decision framework without notes — 8 scenarios, 8 correct answers

🎉 WEEK 3 COMPLETE — NETWORKING & VPC

NEXT: WEEK 3 SATURDAY LIVE LAB